

4 Adult stem cells are undifferentiated cells that are found in most animal tissues.

Adult stem cells can divide by mitosis throughout their lifespan to form identical stem cells (self-renewal) or to form cells that can differentiate into the functioning cells of that tissue.

(a) Mitosis is important for the repair of tissues.

Explain what is meant by repair of tissues.

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..... [1]

(b) Uncontrolled cell division is a characteristic feature of tumour formation from a differentiated cell.

Describe **other** features of tumour formation from a fully differentiated cell.

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..... [2]

(c) Telomeres prevent loss of genes.

Adult stem cells have chromosomes with long telomeres.

Explain why long telomeres are an advantage to cells that carry out many cell cycles.

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..... [2]

[illegible]

Key



- [3]



- (e) Fig. 4.1 shows that monocytes differentiate into cell type **X**, which has a similar function to neutrophils.

Name cell type **X**.

..... [1]

- (f) Cell type **Y** shown in Fig. 4.1 releases molecules with antigen binding sites.

Name the molecules released by cell type **Y**.

..... [1]

- (g) The differentiation of T-lymphocytes begins in the bone marrow and continues in an organ known as the thymus to produce fully differentiated T-helper and T-killer cells.

In the thymus, T-lymphocytes that bind to self antigens are destroyed.

Explain why T-lymphocytes that bind to self antigens need to be destroyed in the thymus.

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..... [3]

[Total: 13]